



# Exploring the importance of floral resources and functional trait compatibility for maintaining bee fauna in tropical agricultural landscapes

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Received: 12 March 2019 / Accepted: 20 February 2020 / Published online: 25 February 2020  
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## Abstract

Loss of semi-natural habitats has been identified as a major driver for pollinator loss in agricultural landscapes. The challenge has been addressed in developed countries by creating flower strips and pollinator habitats in agricultural areas, but this approach has largely been overlooked in developing countries. However, even in countries that adopted pollinator-friendly measures, the complementarities between functional traits of plants and bees are not generally used as selection criteria for candidate species in floral strips. Furthermore, the combined effect of habitat variables that are likely to maximize bee visitation are also overlooked. In tropical developing countries, even the basic information to inform decision making is missing. This paper attempts to bridge this critical gap and reports how assemblages of different non-crop plants could be used to attract bees in tropical small landholdings by investigating non-crop plant identity, non-crop floral traits, bee traits and influential habitat variables at patch scale. Results showed that a combination of both non-crop flower density and diversity increased bee visitation at patch scale. Moreover, trait-based analyses revealed that bees with a larger inter-tegular distance, longer tongue frequently visited flowers with a longer corolla. This study demonstrates that both characteristics of non-crop flower patches and floral traits are important for effective management of non-crops in tropical farmland to attract pollinators with complementary functional traits. We have explored a range of non-crop plants that could be effectively sown to attract bees but recognise that more research is necessary in order to standardise their propagation, establishment and management techniques.

**Keywords** Bee traits · Flower traits · Habitat variables · Non-crop plants · Semi-natural habitat · Tropics

**Electronic supplementary material** The online version of this article (<https://doi.org/10.1007/s10841-020-00225-3>) contains supplementary material, which is available to authorized users.

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## Introduction

The role of semi-natural habitat in supporting wild pollinator fauna has been well documented and has received increased attention as evidence that pollinators are threatened globally has grown (Potts et al. 2010; Vanbergen and The Insect Pollinators Initiative 2013). In agricultural landscapes,

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semi-natural habitats provide both nesting opportunities and diverse food resources to support wild pollinators outside the crop flowering period (Pywell et al. 2005; Russo et al. 2013). Resource distribution and connectivity among various semi-natural habitat patches across a landscape enable the production and survival of wild pollinators (Foley et al. 2005; Williams and Kremen 2007).

Compared to some developed countries in temperate regions, where agricultural landscapes are more homogeneous, agricultural landscapes in the tropics are heterogeneous with diverse landscape elements including patches of semi-natural vegetation closely interspersed with farmlands. Although in tropical countries, agricultural landscapes dominated by small landholdings are rich in biodiversity (Zhou et al. 2018) and thus can potentially sustain healthy communities of beneficial insects, the semi-natural habitat patches have been disappearing fast due to agricultural intensification. Disappearance of tropical habitats has accelerated over the last four decades (DeFries and Rosenzweig 2010; Basu et al. 2016), and the deteriorating state of tropical agricultural landscapes may severely threaten ecosystem services, especially pollination, in the near future.

In most tropical developing countries, creation of floral strips or suitable habitats for pollinators is not part of any agricultural land-management strategy although there is growing awareness regarding the need for such restorative measures. A list of candidate non-crop wild plant species for use in habitat creation schemes would be a useful baseline for habitat restoration projects in the tropics. This is underpinned by the fact that although there is sufficient information on the candidate species for habitat restoration exists for most temperate countries, clearly this information is not directly transferable to the tropics. This is specifically because distinctly different tropical climates and highly mosaic nature of the landscapes would impact flowering phenology, pollinator activity periods, pollinator movements and functional groups of plants and pollinators in different ways than the temperate climatic regions. However, most available information on the subject are from the temperate countries (Hipólito et al. 2018; Tangtorwongsakul et al. 2018; Cusser et al. 2019; Guenat et al. 2019; Riojas-López et al. 2019) and there are no existing prescriptions for the optimal composition of floral strips for pollinators for most tropical countries and definitely not for India. There are also no reports of pollinator-friendly semi-natural habitat creation in tropical agricultural landscapes, including in India. Therefore, lists of candidate non-crop plants for habitat restoration, in different tropical landscapes, are urgently required before any step towards restoration of agricultural landscape are adopted therein.

Even in countries where floral strips or new habitats are created for pollinators (and bees in particular), the selection of candidate plant species is not frequently based upon

functional trait complementarity between flowers and the bee species. Floral traits have been reported to influence plant-pollinator interaction (Irwin et al. 2018). Bees' preferences towards plants have been shown to be related to specific traits such as petal colour, corolla width, corolla length and nectar quality (Gumbert 2000; Gómez et al. 2008; Bauer et al. 2017; Córdoba and Cocucci 2017; Prasifka et al. 2018) or floral display (Eckhart 1991; Mitchell et al. 2004). Besides, specific bee traits (such as length of mouthparts) might have matching complementarity with various floral traits (Maad and Alexandersson 2004; Benitez-Vieyra et al. 2006; Garibaldi et al. 2015). We suggest that the availability of diverse floral traits would ensure visitation by a diverse and more abundant bee fauna with diverse morphological traits catering to a diversity of floral types (of both crop and non-crop plants). Such complementarity can help inform strategies for the creation of floral strips that will attract a diverse bee community to agricultural fields.

The efficiency of diversified floral patches or floral strips to attract diverse bee community will not only depend on their actual composition but also on other habitat variables like floral density, plant density etc. which are little studied. Additionally, studies addressing the combined effect of different patch-scale habitat variables on pollinator visitation in non-crop floral patches are either missing or report contrasting information with respect to these variables (e.g. plant density, plant diversity or floral density of herbs and shrubs) (Hegland and Boeke 2006; Elliott and Irwin 2009). There are several other studies which showcased singular effects of various habitat variables (Tuell et al. 2008; Russo et al. 2013; Campbell et al. 2017). However, none of the above studies investigated combinations of different habitat variables and their role on bee visitation and not in the context of any tropical landscape.

In tropical countries including India, importance of bees and other insects in pollination of important crops and several non-crops including forest trees have been recoded (Hanh et al. 2014; Chakraborty et al. 2018; Saunders and Rader 2019; Barth et al. 2020). However, decline in pollinator populations in tropical heterogeneous landscapes are also evidenced from different studies (Basu et al. 2016; Potts et al. 2016). The vegetation in Indian landscape is hugely diverse ranging from alpine and sub-alpine to arid and semi-arid to tropical evergreen, semi-evergreen or deciduous plant species interspersed with small to medium scale agricultural landholdings and other anthropogenic structures. Over the last four and a half decades, India has undergone large-scale agricultural intensification that involved land conversion to agriculture and consequent landscape simplification through loss of semi-natural habitats (Basu et al. 2011). There is an urgent need to establish an appropriate evidence base to help develop pollinator habitat restoration strategies for the country. To challenge this exploration, this study was conducted

in a region of eastern India where the farming landscapes are facing rapid agricultural intensification and urbanization.

This study, therefore, attempted to answer: (1) Which habitat variables are important in maximising bee visitation at patches of non-crop plants? (2) Which non-crop floral traits attract maximum bee visits? (3) If bee–non-crop floral traits compatibility influence bee visitation.

## Methods

### Study sites

The study area was situated in Balasore, Odisha, an Eastern Indian state. Balasore is a coastal district which is broadly divided into three geographical regions—coastal belt, inner alluvial plain and North-Western hills (Chakraborty 2011; Mahapatra et al. 2012). Our study region falls in the Nilagiri block of the district which is situated in the North-Western hilly region (Chakraborty 2011; Mahapatra et al. 2012). The landscape is a mosaic of agricultural fields, unmanaged hedgerows, semi-natural habitats and sparse deciduous and evergreen tree patches. The soil is mostly red and yellow lateritic type, mixed with gravels. The annual average temperature is 32 °C and annual average rainfall is around 1300 mm (Mahapatra et al. 2012). Study region is broadly mosaic of farmlands, man-made structures as well as several randomly situated semi-natural habitat patches. In the overall hot and dry climate of study region, non-crop plant flowering peaks during monsoon to post-monsoon and then gradually declines through winter. Hence, the study was conducted between July 2014 and January 2015 (monsoon to winter).

A total of 20 study sites were selected randomly within all the available semi-natural habitats across a broad landscape covering an area of about 728 km<sup>2</sup> (from 21° 35' 6.89" N, 86° 32' 6.86" E to 21° 21' 17.06" N, 86° 48' 35.34" E) (Online Resource 1). These semi-natural habitats are more or less surrounded by agricultural fields where a variety of crops (for example, eggplant (*Solanum melongena*), cucumber (*Cucumis sativus*), pumpkin (*Cucurbita maxima*), chilli (*Capsicum frutescens*) and rice (*Oryza sativa*)) are cultivated throughout the year. The average distance between the sites was 10.68 km (SE ± 0.39). All the study sites were situated on plain lands having more or less uniform inclination and altitude (74.3 m; SE ± 6.06).

### Bee visit observation

The study sites varied in size and vegetation composition. At each study site, one 50 m × 50 m permanent plot was established in which three permanent 10 m × 10 m plots were located randomly. All observations were taken within those 50 m × 50 m established plots. In each 10 m × 10 m plot,

three permanent 1 m × 1 m quadrats were randomly selected to carry out focal observation of bee visitation and estimation of habitat variables. To minimize the temporal variation in observation among the sites, sampling was repeated three times through the study period in each study site with an interval of average 22 days. At each study site, observation of bee visitation was carried out for 10 min session in each 1 m × 1 m plot from 0700 to 1500 h (i.e. total 24 h of observation per study site). A 15 min interval was taken between two consecutive focal observations. During each focal observation, all the bees visiting and touching the reproductive parts of the non-crop flowers present in 1 m × 1 m quadrats were recorded and respective flower species were noted down for further analysis. Bees were collected by sweep net and were put into a killing jar (with ethyl acetate). Collected bee specimens were dried and pinned for preservation (Gullan and Cranston 2014) and taxonomic identification. Observations were only carried out on sunny dry days.

### Measurement of habitat variables

The number of non-crop plants (herbs and shrubs) and their species identity were recorded within each 1 m × 1 m quadrat in each study site. To minimize the effect of temporal variation in non-crop plant density within each site, vegetation sampling was performed twice during the first and third day of focal observation. Number of open flowers (for an estimation of flower density) for each non-crop species was also recorded (Hamblin et al. 2018) once during each day of focal observation. Plants with Girth at breast height (GBH) > 10 cm and height ≥ 5 m were considered to be trees (Gomes et al. 2008). Trees were not included in the bee visitation survey. However, to take into account the probable effect of flowering trees on bee abundance, the total number of trees in flower (NOTF) and total number of tree species in flower (NOTSF) were counted on each day of focal observation within each 10 m × 10 m quadrat at each site. Non-crop plant species and leaves of each tree species were collected and preserved in a herbarium for further taxonomic identification. Trees were also photographed during blooming (if they bloomed at all during the study period) and further classification was done at lowest possible taxonomic category.

### Bee trait measurements

We conducted morphological measurements on all bee individuals collected during focal observations. For each bee species, 1 to 10 individuals (depending on availability) were measured in the laboratory after preservation. Bee body length (BOL) (from the front of the head to the tip of the abdomen), inter-tegular distance (ITD; the length

between the two originating points (tegulae) of wings) and lengths of foreleg (FLL) and midleg (MLL) (from base of femur to tip of tarsal claws) were measured. For estimation of tongue (proboscis) length (TOL), we measured length from the basal part of the tongue (i.e., from labiomaxillary tube, where it attaches to the head) to the tip of the glossa after the tongue was fully stretched. When necessary, bee's head was dissected for tongue measurement. All the traits were measured with a stereomicroscope (Olympus SZ61) and ocular micrometer scale. Finally, all the measurements in ocular divisions were converted to micrometer and subsequently to millimetre (Online Resource 2).

### Non-crop (herbs and shrubs) floral trait measurements

Flowers of non-crops were collected for the measurements of floral traits from same 1 m × 1 m quadrats where the bee visitation study was conducted. Flower traits were measured for 15 individual flowers of each species. Corolla length (LEN), corolla opening diameter (DIA) and sugar content (SUG) in the nectar were measured for each non-crop flower species visited by bees. Corolla length and diameter were measured by Vernier callipers. Corolla length was measured as the length from the base of the calyx to the longest end of the petal (Engel and Irwin 2003). Corolla diameter was measured as the longest width of the corolla opening (Engel and Irwin 2003). For Asteraceae flowers, we measured floret cluster diameter as corolla diameter and floret length as corolla length. Corolla colour was also noted for every bee visited plant species and further categorised into following categories: blue-violet flower; reddish flower; white flower; white combination flower; yellow flower. These colour categories were made based on the human colour perception and may further include other colours that are easily distinguished by the bees (for example, yellow coloured corolla with or without UV reflecting property). Nectar was extracted following standard protocols using glass capillary tubes (size 1 µl microcaps®, Drummond Scientific) (Corbet 2003; Phillips et al. 2018) and its sugar percentage was measured using a regular hand-held graduated refractometer (ERMA Inc.) (Phillips et al. 2018). For each non-crop species, nectar was extracted from individual flowers until 5 µl nectar was collected. It was then mixed well with 100 µl distilled water. The solution was then placed on the refractometer and the Brix % value was recorded. This process was repeated 15 times for each non-crop species to minimize uncertainty in measuring error; mean value was used for analysis. The refractometer was calibrated with distilled water at the field weather condition to minimize the error prior to the sampling.

### Data analyses

For each of the 20 sites, number of open non-crop flowers and number of non-crop plants, recorded from nine 1 m × 1 m quadrats, were averaged to estimate non-crop flower density (FD) and non-crop plant density (NCD) which were included in the model. For each study site, the bee visitation rate (VR) was estimated by averaging the number of individual bees visiting each 1 m × 1 m quadrat across all observation sessions across all observational days. Total number of trees in flower (NOTF), total number of tree species in flower (NOTSF), non-crop flower diversity (FDIV), and non-crop flower colour diversity (CDIV, number of flowers present in each colour category) were also calculated for each site. NOTF and NOTSF were calculated by summing up the total number of individual trees having flowers and the total number of tree species having flowers in three 10 m × 10 m plots in each site, respectively. FDIV and CDIV were estimated using the Shannon and Weiners' (1949) equation

$$FDIV = - \sum_{i=1}^S p_i \cdot \ln p_i$$

where 'S' was the number of non-crop plant species and 'p<sub>i</sub>', the proportion of flowers in each 'ith' non-crop species.

Effects of non-crop flower density (FD), non-crop flower diversity (FDIV), non-crop flower colour diversity (CDIV), total number of trees in flower (NOTF), total number of tree species in flower (NOTSF) and non-crop plant density (NCD) on bee visitation rate (VR) were analysed using a generalized linear model (GLM) where FD, FDIV, CDIV, NOTF, NOTSF, NCD were predictor variables and VR the response. We fitted our model with different error functions e.g. 'inverse.gaussian', 'gaussian' and 'Gamma' with appropriate link function e.g. 'log', 'identity' and 'inverse'. All the combinations of predictor variables were used in the model. Models were first ranked according to their respective AICc (Akaike's information criterion corrected for small sample size) value and respective AICc weight (w<sub>i</sub>). Generally, models with 'ΔAICc > 2' were considered to be less supportive models (Burnham and Anderson 2003). The best approximating models were determined after creating a 95% confidence interval according to the 'Information-theoretic approaches' proposed by Burnham and Anderson (2003). Model averaging was also conducted to visualize the effect of different predictor variables in the model (Burnham and Anderson 2003).

Relations between corolla length (LEN), corolla diameter (DIA) and nectar sugar percentage (SUG) with bee visitation rate (VR) were also explored by generalized linear regression (GLM) method where LEN, DIA and SUG were predictor variables and the VR was response. Residual errors for



these three separate regressions were different and therefore, the combination effect of predictor variables could not be explored through ‘Information-theoretic approaches’ proposed by Burnham and Anderson (2003).

Morphological trait data (ITD and TOL) of those bee species that made visits to the non-crop flowers were arranged sequentially according to the data of floral traits (LEN and DIA) of those non-crop species which were visited by the bees. Relations between bee traits (ITD and TOL as response variables) and floral traits (LEN, DIA and SUG as predictor variables) were, then, explored by generalized linear regressions (GLM). Bee traits that were multicollinear with each other were not included in the regression (Online Resource 3).

All the plots were created from GLM analyses, where a sequence of length values was first set up for predictor variable and fitted values (for response variable) were, then, predicted within the sequence of length values from respective GLM analyses.

Normality and heteroscedasticity were checked by normal Q-Q plots and standardized residuals against fitted value plots (Crawly 2007). Variance inflation factor (VIF) tests were performed to check the multicollinearity among the predictor variables. ‘VIF’ value  $> 5$  was considered as multicollinear (Craney and Surles 2002; Akinwande et al. 2015; Vu et al. 2015) (Online Resource 3 and 4). Non-parametric tests were performed in all the cases as our data did not meet normality even with transformation. All the analyses were performed in ‘R’ statistical software (version 3.4.3) (R Core Team 2013) with MuMIn (Barton 2017), AICcmodavg (Mazerolle 2017), and ‘fmsb’ (Nakazawa 2017) packages.

## Results

About 15,000 non-crop plant individuals (herbs and shrubs together) were recorded across all of the 20 study sites. These belonged to a total of 78 species from 32 families and 59 genera. Six non-crop species could not be identified at any taxonomic level (Table 1). Family Asteraceae had 7 representative species which is the highest in number in a single-family followed by Acanthaceae (6 species), Fabaceae (6 species), Rubiaceae (6 species), Lamiaceae (5 species), Caesalpiniaceae (4 species), Euphorbiaceae (4 species) and Malvaceae (4 species). The rest of plant families were represented by one or two plant species.

Among this wide range of non-crop plant species, only a total of 25 species (32.05%) belonging to 12 genera were observed being visited by bees during our study (Table 1). Total number of bee-visited non-crop species under each plant family was shown in Fig. 1. Although non-crops belonging to Fabaceae, Lamiaceae, Acanthaceae and Caesalpiniaceae attracted higher number of bees, there was no

significant difference in bee visitation across non-crop plant families (Kruskal–Wallis test,  $df = 12$ ,  $p > 0.05$ ).

A total of 146 individuals belonging to 26 different tree species (Online Resource 5 and 6) were also recorded at the study sites.

A total of 293 bee individuals, from 3 families and 22 species, were recorded during the period of our study. Of the 22 species, 82.9% of individuals belonging to 20 species touched the floral reproductive parts during visit.

## Effects of habitat variables on bee visitation

Variation in wild bee visitation rate was explained by five best candidate models which were within 95% confidence interval (Table 2). Out of these candidate models, ‘FD + FDIV’ (non-crop flower density + non-crop flower diversity) and ‘FD + FDIV + NOTF’ (non-crop flower density + non-crop flower diversity + total number of trees in flower) were the best parsimonious models which determined bee visitation. However, ‘FD + FDIV’ had greater ‘Akaike weight’ ( $w_i = 0.50$ ) than ‘FD + FDIV + NOTF’ ( $w_i = 0.22$ ). All the combination models are listed in Online Resource 7. FD was the most important factor in the models followed by FDIV as ascertained by “Model-averaging of parameters” (Table 3). Although, NOTF was one of the variables selected in the best parsimonious models, it had negligible ‘Relative Importance’ compared to the other two variables in the models (Table 3).

## Relationship between floral traits and bee visitation

Bee visitation rate (VR) was significantly determined by nectar sugar content, corolla length and corolla diameter. Visitation rate significantly increased with increase in nectar sugar content ( $t = 11.14$ ,  $p < 0.0001$ ) and corolla length ( $t = 4.13$ ,  $p < 0.001$ ) (Fig. 2a, b). However, visitation rate dropped with increase in corolla diameter ( $t = -2.25$ ,  $p = 0.03$ ) (Fig. 2c). Although corolla diameter had significant negative relationship with nectar sugar content ( $t = -3.735$ ,  $p < 0.01$ ) (Fig. 3).

## Relationship between floral traits and bee traits

All the measured floral traits are listed in Online Resource 8. Bees with larger ITD visited flowers having larger corolla diameter ( $t = -2.157$ ,  $p = 0.0348$ ) and longer corolla length ( $t = -3.139$ ,  $p = 0.00256$ ) significantly more frequently (Fig. 4a, b). Also, we found significant relationship between bee tongue length (TOL) and corolla length of flowers ( $t = -3.698$ ,  $p = 0.000454$ ) (Fig. 4d). However, tongue length and corolla diameter did not show any significant relationship ( $t = -1.838$ ,  $p = 0.0707$ ) (Fig. 4c). Also, there were no significant relationships among tongue length

**Table 1** List of herb and shrub species encountered during focal observation in the study sites

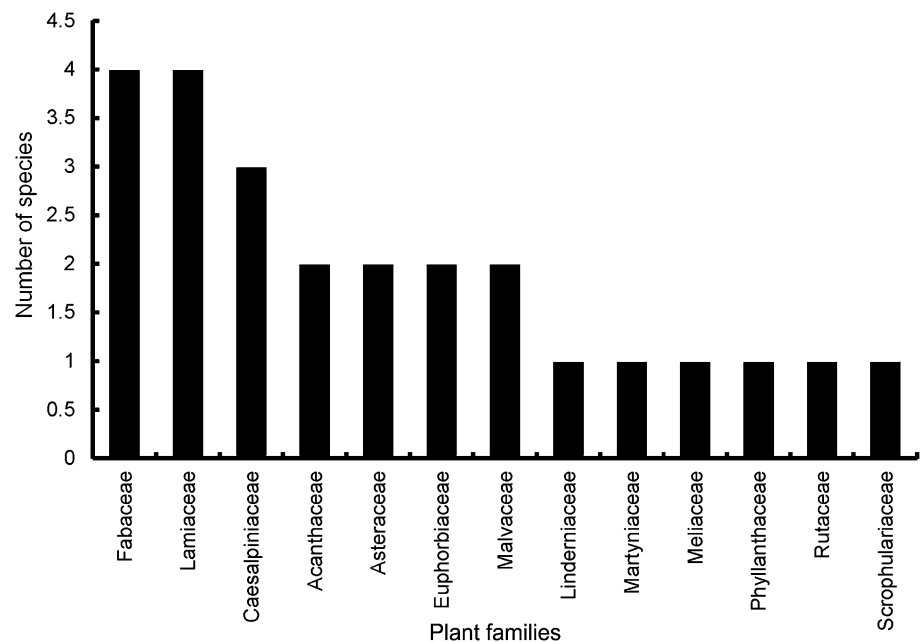
| Serial no | Binomial name                                                    | Family           |
|-----------|------------------------------------------------------------------|------------------|
| 1         | <i>*Rungia parviflora</i>                                        | Acanthaceae      |
| 2         | <i>Clerodendrum viscosum</i> (= <i>C. infortunatum</i> )         | Lamiaceae        |
| 3         | <i>Holarrhena pubescens</i>                                      | Apocynaceae      |
| 4         | <i>*Sida veronicaefolia</i> (= <i>S. humilis</i> )               | Malvaceae        |
| 5         | <i>Croton oblongifolius</i>                                      | Euphorbiaceae    |
| 6         | <i>*Desmodium triflorum</i>                                      | Fabaceae         |
| 7         | <i>Meyna spinosa</i> ( <i>Xeromphis</i> )                        | Rubiaceae        |
| 8         | <i>Emilia sonchifolia</i>                                        | Asteraceae       |
| 9         | <i>Phyllanthus simplex</i>                                       | Phyllanthaceae   |
| 10        | <i>Spermacoce hispida</i>                                        | Rubiaceae        |
| 11        | <i>Evolvulus nummularius</i>                                     | Convolvulaceae   |
| 12        | <i>Hybanthus enneaspermus</i> (= <i>Ionidium suffruticosum</i> ) | Violaceae        |
| 13        | <i>*Sida acuta</i>                                               | Malvaceae        |
| 14        | <i>Ludwigia parviflora</i>                                       | Onagraceae       |
| 15        | <i>Sida rhombifolia</i> var. <i>rhomboidea</i>                   | Malvaceae        |
| 16        | <i>Melochia corchorifolia</i>                                    | Sterculiaceae    |
| 17        | <i>Corchorus aestuans</i>                                        | Tiliaceae        |
| 18        | <i>*Hyptis suaveolens</i>                                        | Lamiaceae        |
| 19        | <i>Didelacanthus prostratus</i> (= <i>Ruellia prostrata</i> )    | Acanthaceae      |
| 20        | <i>Elephantopus scaber</i>                                       | Asteraceae       |
| 21        | <i>Hedyotis pinifolia</i>                                        | Rubiaceae        |
| 22        | Unidentified plant 1                                             | —                |
| 23        | <i>Bonnaya brachiata</i> (= <i>Lindernia ciliata</i> )           | Linderniaceae    |
| 24        | <i>Evolvulus alsinoides</i>                                      | Convolvulaceae   |
| 25        | <i>*Glycosmis pentaphylla</i>                                    | Rutaceae         |
| 26        | <i>Dicliptera</i> sp.                                            | Acanthaceae      |
| 27        | <i>Vernonia cinerea</i>                                          | Asteraceae       |
| 28        | <i>Oldenlandia corymbosa</i>                                     | Rubiaceae        |
| 29        | <i>Alternanthera sessilis</i>                                    | Amaranthaceae    |
| 30        | <i>Lobelia</i> sp.                                               | Campanulaceae    |
| 31        | <i>Spermacoce ocyroides</i>                                      | Rubiaceae        |
| 32        | <i>Scoparia dulcis</i>                                           | Plantaginaceae   |
| 33        | Unidentified plant 2                                             | —                |
| 34        | <i>*Anisochilus carnosus</i>                                     | Lamiaceae        |
| 35        | Unidentified plant 3                                             | —                |
| 36        | Unidentified plant 4                                             | —                |
| 37        | <i>Taraxacum officinale</i>                                      | Fabaceae         |
| 38        | <i>Biophytum sensitivum</i>                                      | Oxalidaceae      |
| 39        | <i>*Limnophila repens</i>                                        | Scrophulariaceae |
| 40        | <i>Hydrolea zeylanica</i>                                        | Hydroleaceae     |
| 41        | <i>Oldenlandia</i> sp.                                           | Rubiaceae        |
| 42        | <i>Erythraea roxburghii</i>                                      | Gentianaceae     |
| 43        | <i>Drosera burmannii</i>                                         | Droseraceae      |
| 44        | <i>Eranthemum</i> sp.                                            | Acanthaceae      |
| 45        | <i>Barleria prionitis</i>                                        | Acanthaceae      |
| 46        | <i>Ageratum conyzoides</i>                                       | Asteraceae       |
| 47        | <i>Boerhavia repens</i>                                          | Nyctaginaceae    |
| 48        | <i>Crotalaria prostrata</i>                                      | Fabaceae         |
| 49        | <i>*Lindernia crustacea</i>                                      | Linderniaceae    |
| 50        | Unidentified plant 5                                             | —                |
| 51        | <i>*Phyllanthus rheedei</i>                                      | Phyllanthaceae   |

**Table 1** (continued)

| Serial no | Binomial name                                                | Family           |
|-----------|--------------------------------------------------------------|------------------|
| 52        | <i>Allophylus cobbe</i>                                      | Sapindaceae      |
| 53        | <i>Cleome viscosa</i>                                        | Cleomaceae       |
| 54        | * <i>Ocimum canum</i>                                        | Lamiaceae        |
| 55        | * <i>Ocimum basilicum</i>                                    | Lamiaceae        |
| 56        | <i>Mollugo stricta</i>                                       | Molluginaceae    |
| 57        | Unidentified plant 6                                         | –                |
| 58        | * <i>Crotalaria pallida</i>                                  | Fabaceae         |
| 59        | * <i>Cipadessa baccifera</i>                                 | Meliaceae        |
| 60        | <i>Adenosma indianum</i>                                     | Scrophulariaceae |
| 61        | <i>Urena lobata</i>                                          | Malvaceae        |
| 62        | <i>Spilanthes paniculata</i>                                 | Asteraceae       |
| 63        | <i>Achyranthes bidentata</i>                                 | Amaranthaceae    |
| 64        | <i>Jatropha curcas</i>                                       | Euphorbiaceae    |
| 65        | * <i>Stylosanthes hamata</i>                                 | Fabaceae         |
| 66        | * <i>Cassia occidentalis</i>                                 | Caesalpiniaceae  |
| 67        | * <i>Cassia tora</i>                                         | Caesalpiniaceae  |
| 68        | * <i>Cassia alata</i>                                        | Caesalpiniaceae  |
| 69        | * <i>Tephrosia purpurea</i>                                  | Fabaceae         |
| 70        | * <i>Tridax procumbens</i>                                   | Asteraceae       |
| 71        | <i>Mimosa pudica</i>                                         | Mimosaceae       |
| 72        | * <i>Chromolaena odorata</i> (= <i>Eupatorium odoratum</i> ) | Asteraceae       |
| 73        | * <i>Jatropha gossypifolia</i>                               | Euphorbiaceae    |
| 74        | <i>Bryophyllum</i> sp.                                       | Crassulaceae     |
| 75        | * <i>Andrographis paniculata</i>                             | Acanthaceae      |
| 76        | <i>Cassia hirsuta</i>                                        | Caesalpiniaceae  |
| 77        | * <i>Croton bonplandianus</i>                                | Euphorbiaceae    |
| 78        | * <i>Martynia annua</i>                                      | Martyniaceae     |

Asterisk (\*) indicates species which received bee visits during focal observation

**Fig. 1** Family wise number of plant species visited by bees. Horizontal axis represents number of plant families which were observed to be visited by bees. Height of the column represents number of plant species (under each respective family) that were visited by bees during the study period



**Table 2** Best candidate models (according to 95% confidence interval) showing effects of different habitat variables (FD=non-crop flower density, FDIV=non-crop flower diversity, NCD=non-crop plant density, NOTF=total number of trees in flower and NOTSF=total number of tree species in flower) on bee visitation rate (VR)

| Response | Predictor               | AIC <sub>c</sub> | ΔAIC <sub>c</sub> | wi          |
|----------|-------------------------|------------------|-------------------|-------------|
| VR       | <b>FD + FDIV</b>        | <b>− 46.75</b>   | <b>0</b>          | <b>0.50</b> |
|          | <b>FD + FDIV + NOTF</b> | <b>− 45.14</b>   | <b>1.61</b>       | <b>0.22</b> |
|          | FD                      | − 44.30          | 2.46              | 0.15        |
|          | FD + FDIV + NCD         | − 43.15          | 3.61              | 0.08        |
|          | FD + NCD                | − 41.78          | 4.97              | 0.04        |

The best models are shown in bold

NOTSF was multicollinear with NOTF and therefore not included in the analysis

(TOL) ( $t=0.011$ ,  $p=0.992$ ) and ITD ( $t=0.012$ ,  $p=0.991$ ) with nectar sugar percentage (SUG) found in our study.

## Discussion

Carefully designed non-crop flower mixtures for flowering strips/margins can help improve the capacity of intensively managed landscapes to maintain bee diversity (Kremen et al. 2004; Morandin and Kremen 2013). For effective delivery of pollination services in order to enhance crop yield, it is important to maximize flower visits by attracting the optimum number of bees (Geslin et al. 2017). Given the limited space available for creation of bee attracting plant habitat patches in a tropical agricultural landscape, it is especially crucial to select appropriate plant species that maximise bee visits. Only about one-third of the non-crop plant species recorded from the studied agricultural landscape attracted bees and these belonged to a restricted subset of 13 plant families. This shows the narrow choice available for selection of candidate non-crop plants that have potential to create bee-friendly habitats in our study area.

In our study, non-crop plant species belonging to family Fabaceae and Lamiaceae received higher number of bees than other families. Fabaceae flower morphology has evolved to maximize bee pollination (Aronne et al. 2012). The aestivation of Fabaceae flowers is vexillary type (closed type flower) where a vexillum or banner petal protrudes from the base of the calyx while the other petals cover the reproductive parts (Aronne et al. 2012). Banner petals generally have nectar guides that attract pollinators (Aronne et al. 2012) and guide them to the nectar source. The second most important group in our study was Lamiaceae which consists of 16% of the total plant species visited by bees. Aestivation of Lamiaceae flowers are, generally, imbricate or valvate (open type flower) where both pollen and nectar are easily accessible to pollinators (Harley et al. 2004). Although, several species of Lamiaceae, due to their morphological constraints, can be restrictive for different insect flower visitors, the Lamiaceae flowers in our study region had smaller corolla tubes with exposed reproductive parts and therefore did not pose any constraints for the visitors. Lamiaceae flowers generally consist of five petals where three petal lobes are fused to form the largest petal structure (Elpel 2000). The petals also consist of nectar guides which attract pollinators to the flower (Harley et al. 2004). Our study suggests that Fabaceae and Lamiaceae should be essential components of seed mixtures for creation of flower-rich patches in the agricultural landscape in our study region to attract bees. Nevertheless, further studies are required to identify more bee attracting non-crop plant species and their phenotypic traits in other unexplored parts of India and other tropical agricultural landscapes.

Patch characteristics (or variables) of the local habitats are also important in attracting bee species for pollination (Williams and Winfree 2013). Whether density or diversity of non-crop flowers is more important in attracting a higher number of bees has been a matter of debate (Basu et al. 2016; Rollin et al. 2019). As the study shows, among all the combinations, the combined effect of both non-crop flower density and diversity is more important in attracting a higher number of bees. Flower density in our study was found to be

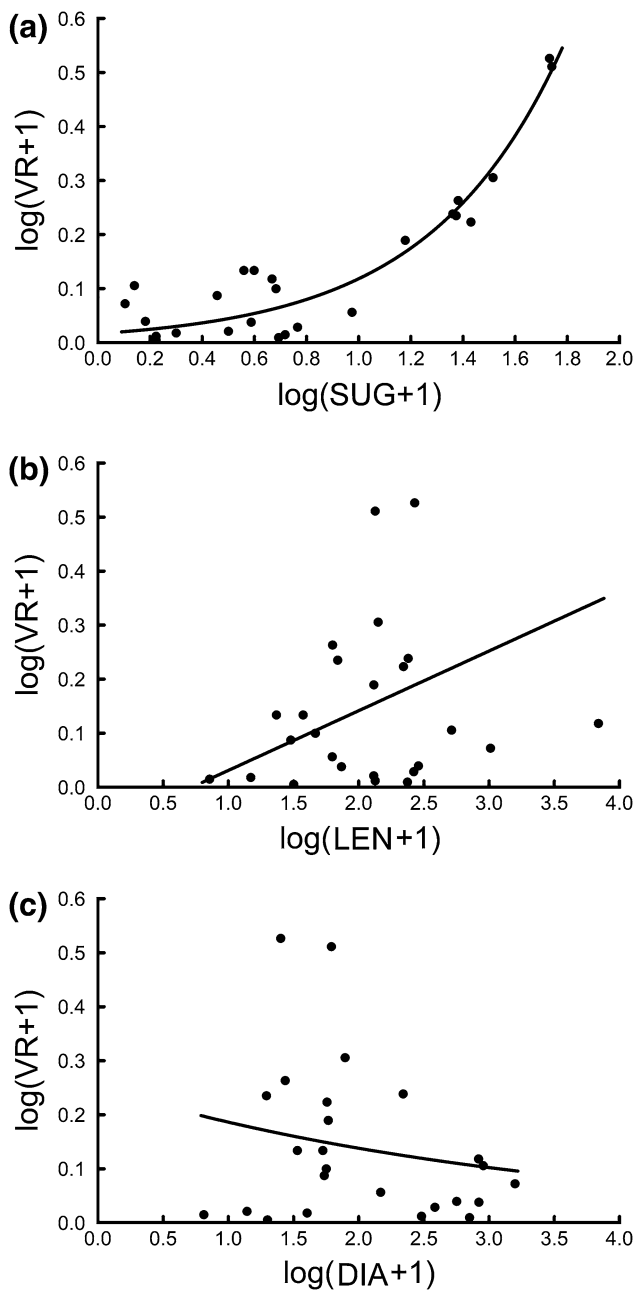
**Table 3** Summary of model averaging (GLM) analysis and relative importance of each habitat variable (FD=non-crop flower density, FDIV=non-crop flower diversity, NCD=non-crop plant density and

NOTF=total number of trees in flower) while response variable was bee visitation rate (VR)

| Response | Habitat variables | Model-averaged coefficients | Standard error (adjusted) | z     | p      | Relative importance |
|----------|-------------------|-----------------------------|---------------------------|-------|--------|---------------------|
| VR       | FD                | 0.0028877                   | 0.0006040                 | 4.781 | <0.001 | 1                   |
|          | FDIV              | 0.0598033                   | 0.0271557                 | 2.202 | <0.05  | 0.81                |
|          | NOTF              | 0.0015741                   | 0.0013755                 | 1.144 | 0.25   | 0.22                |
|          | NCD               | 0.0003819                   | 0.0013614                 | 0.281 | 0.78   | 0.12                |

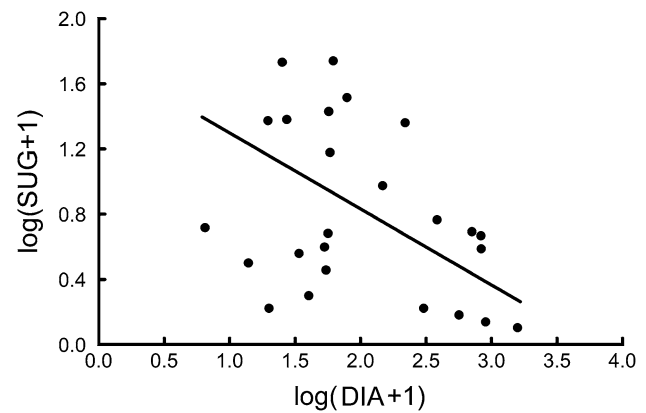
The relative importance =  $\Sigma wi$





**Fig. 2** Relationships between bee visitation rate (VR) and **a** nectar sugar content (SUG), **b** corolla length (LEN) and **c** corolla diameter (DIA). Regression lines were plotted after GLM fitting

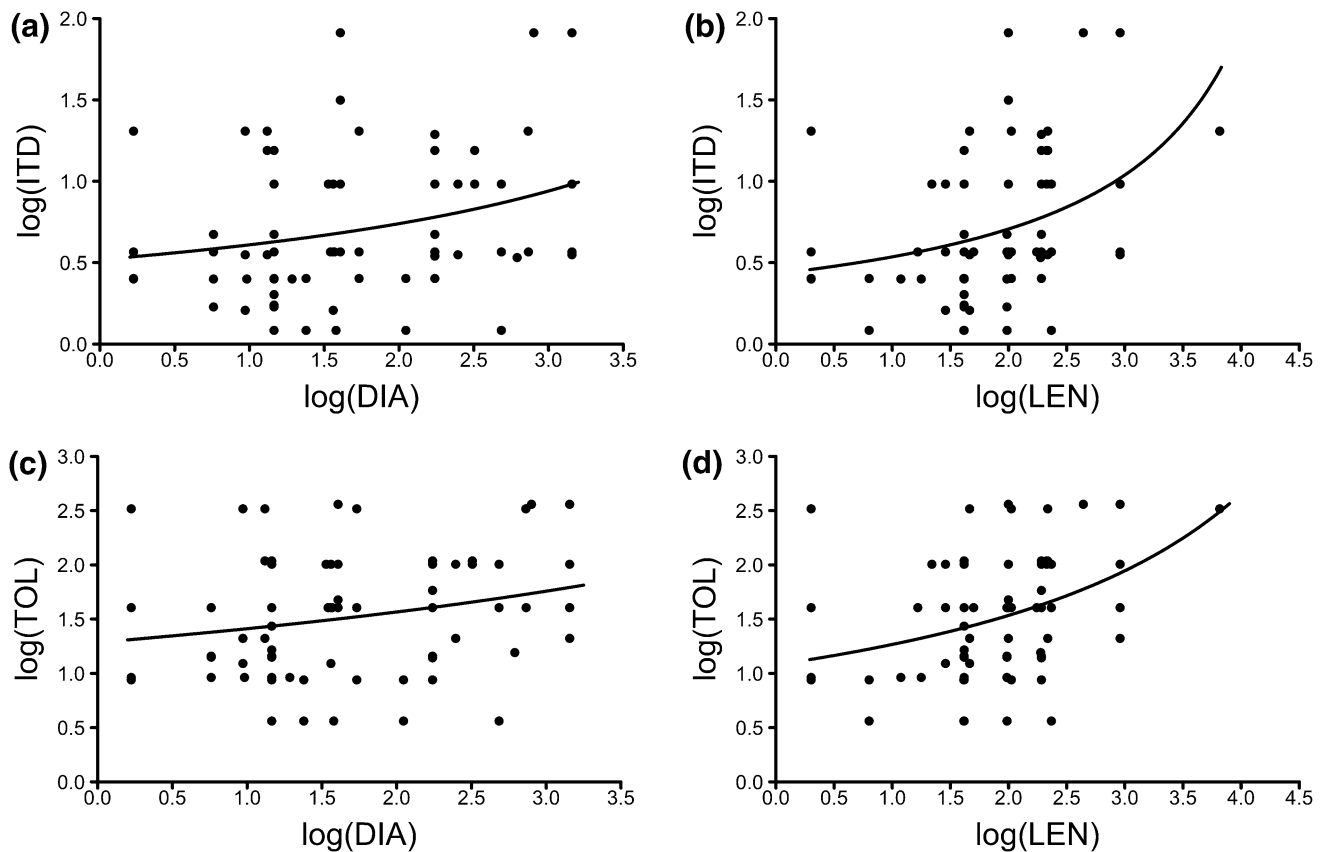
the most important habitat variable and has been previously reported as being a crucial one for increasing bee visitation (Hegland and Boeke 2006). High flower density acts as a visual signal of resource availability to the bees (Dafni et al. 1997). Therefore, in a fragmented landscape, at patch scale, non-crop habitats with high flower density are an attractive foraging ground for pollinators, and subsequently, non-crops and nearby crops might involve in competition for pollinator visitation. However, presence of non-crops



**Fig. 3** Relationship between nectar sugar content (SUG) and corolla diameter (DIA). Regression line was plotted after GLM fitting

was observed to increase the probability of pollination in nearby pollinator-dependent crops (Morandin and Kremen 2013), and this facilitation may be sufficient to minimize the effect of competition for pollinator visitation between non-crops and crops. Flower diversity was the second most important factor that influenced bee visitation to the flowers in our studied non-crop habitat patches. The presence of diversified flowering species increases the resource diversity for the pollinators, thereby attracting a diverse pollinator assemblage (MacArthur 1972; Russo et al. 2013). Flowers of different species of plants contain different nutritional ingredients, including sugars (Nicolson 2011). Different bee species display differential preferences towards nectar sugar (Wykes 1952; Waller 1972; Nicolson 2011) and therefore diverse flower patches with associated differential sugar content, may attract a diverse bee community regulated by their nutritional preferences. Most of the information regarding nectar nutritional ingredients has been studied in the *Apis* species and there remains a considerable knowledge gap regarding nutritional composition and nectar preferences for wild bees.

Diversified flower patches can also increase trait-based compatibility for attracting or sustaining a diversified bee community (Pontin et al. 2006; Campbell et al. 2012). This, in turn, may increase overall bee visitation. Our study revealed the existence of a close relationship between bee visitation and flower traits. A patch with high species diversity may be assumed to have a high trait diversity as well. Availability of diverse floral traits may help to sustain a higher diversity of floral visitors with diverse morphological traits. This explains higher bee visitation towards patches with increasing flower diversity in our study area. As our results reveal, specific floral traits attracted a higher number of bees. For example, visitation seemed to be influenced by corolla length, and corolla diameter as more tubular flowers attracted a higher number



**Fig. 4** Relationships between bee inter-tegular distance (ITD) and tongue length (TOL) with **a, c** corolla diameter (DIA) and **b, d** corolla length (LEN). Regression lines were plotted after GLM fitting

of bees. This means in our study area, presence of tubular flowers should be an essential component of floral strips. Again, while certain bees with compatible morphology and size can access and explore rewards in tubular flowers, visitation from bees that cannot access floral rewards in such flowers would also require availability of non-tubular flowers. Smaller bee species may explore the resource easily by entering the corolla tube and generally provide more visitation to the flowers (Sahli and Conner 2007; Mallinger and Prasifka 2017). Sometimes, body size does not permit larger bees completely/partly to enter the corolla tube and as a consequence, their proboscis cannot reach the nectary. Our results also showed that bee visitation increased with increasing sugar concentration of the nectar. Bees' preference towards higher sugar concentration also have been observed in studies comparing different flower species (Roubik and Buchmann 1984; Bailes et al. 2018; Masierowska et al. 2018). A previous study has shown that sugar type may vary with floral structure and tubular flowers contain more sucrose in nectar (Southwick et al. 1981). This may be the reason for higher bee visitation in tubular flowers since bees have a distinct preference for sucrose compared to fructose and hexose (Biesmeijer et al. 1999).

The positive association of inter-tegular distance and tongue length of foraging bees with corolla length of flowers found in our study is also supported by earlier studies (Bommarco et al. 2012; Klumpers et al. 2019). This also goes with the concept that flower structures seem to have co-evolved with specialized pollinators to increase pollination success by providing structural compatibility to the latter (Herrera 1996; Jauker et al. 2016; Pauw et al. 2017; Woźniak and Sicard 2018). Specialised pollination syndromes can also promote flower constancy, reduce interspecific pollen transfer and therefore provide better reproductive fitness for the plants (Krishna and Keasar 2018). This concept can be applied for the preparation of bee-friendly habitats to enhance insect-flower interactions.

As our study shows, an ideal seed mix for creation of floral strips in our study area that can attract bees in good numbers has to have a bias for species with flowers having long and tubular corolla, preferably belonging to families Fabaceae and Lamiaceae. Further, our study also indicates that such floral strips have to have high plant species diversity with synchronous flowering. However, the trait matching was based on the bee species assemblage available in our study area and the pattern might be different in other

areas where different floral traits may emerge as important. Therefore, we argue that similar studies need to be carried out in other areas of the tropics. Moreover, seasonal studies would also be necessary for understanding the temporal fluctuations of plant-pollinator interactions in semi-natural habitats and therefore, selection of seed mixtures to support pollinator throughout the years will be more appropriate. Our study at this moment is a pioneering attempt (as far as we are aware) to select candidate plant species using a trait-based approach for creating pollinator habitats in the tropics. Hopefully, this study would be a possible starting point that will inspire others to explore more information for conserving or creating such habitats in the tropics.

## Conclusions

Small and fragmented land holdings in the tropics, including India, pose a major constraint in sparing land for creation of pollinator habitat. In most cases, the only land available for such purpose is the meagre area of bunds that divide the landholdings. Creation of pollinator-attracting vegetation in this limited available space would, therefore, entail careful selection of non-crop plants with traits appropriate for maximising bee visits with matching functional traits. Knowledge of non-crop plants that can qualitatively improve the agricultural landscape is a major pre-requisite; the present work is an important contribution towards that in the context of Indian agriculture. However, given the situations in some developed countries where traditionally managed semi-natural habitats have been completely destroyed, our primary goal should be beyond mere creation of floral strips and focussed more towards conservation of natural/semi-natural areas across agricultural landscapes before it is too late.

**Acknowledgements** Our thanks to the farmers and other villagers for providing access to their lands and hospitality at times of need. We also thank Dr. Debabrata Maity, Department of Botany, University of Calcutta and Aditi Dutta for helping in the identification of plant specimens and morphometry of bees, respectively. This project was funded by the Darwin Initiative (project 19-024) and Council of Scientific & Industrial Research (sanction no. 09/028(0882)/2012-EMR-I, date 21.08.2012). We also thank the reviewers for their valuable comments on an earlier draft of the manuscript.

## Compliance with ethical standards

**Conflict of interest** The authors declare no conflict of interest.

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